

Started on	Wednesday, 28 January 2026, 10:30 AM
State	Finished
Completed on	Wednesday, 28 January 2026, 10:45 AM
Time taken	14 mins 56 secs
Marks	12.00/12.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct

Mark 2.00 out of 2.00

A scalar λ is an eigenvalue of an $n \times n$ matrix A if there exists a nonzero vector X such that:

- ☐ A. $AX = 0$
- ☒ B. $AX = \lambda X$ ✓
- ☐ C. $A^T X = \lambda X$
- ☐ D. $(A + \lambda I)X = 0$

Your answer is correct.

The correct answer is: $AX = \lambda X$

Question 2

Correct

Mark 2.00 out of 2.00

The **characteristic equation** of a matrix A is:

- ☐ A. $|A| = 0$
- ☐ B. $|A + \lambda I| = 0$
- ☒ C. $|A - \lambda I| = 0$ ✓
- ☐ D. $|A^T - \lambda I| = 0$

Your answer is correct.

The correct answer is: $|A - \lambda I| = 0$

Question 3

Correct

Mark 2.00 out of 2.00

If A is diagonalizable and $A = PDP^{-1}$, then A^k equals:

- ☐ A. $A^k = P^k D P^{-k}$
- ☒ B. $A^k = P D^k P^{-1}$ ✓
- ☐ C. $A^k = D^k$
- ☐ D. $A^k = P^{-1} D^k P$

Your answer is correct.

The correct answer is: $A^k = P D^k P^{-1}$

Question 4

Correct

Mark 2.00 out of 2.00

Use Cayley–Hamilton to reduce a power

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

which is correct?

- ☒ A. $A^2 - 5A + 6I = 0$ ✓
- ☐ B. $A^2 + 5A + 6I = 0$
- ☐ C. $A^2 - 6A + 5I = 0$
- ☐ D. $A^2 - 5I = 0$

Your answer is correct.

The correct answer is: $A^2 - 5A + 6I = 0$

Question 5

Correct

Mark 2.00 out of 2.00

Suppose A is diagonalizable with eigenpairs

$\lambda_1 = 2, \lambda_2 = 5$ and eigenvectors v_1, v_2 (linearly independent).
Which statement is correct?

- ☐ A. $A^n = (2^n + 5^n)I$
- ☐ B. $A^n v_1 = n2 v_1$ and $A^n v_2 = n5 v_2$
- ☒ C. $A^n v_1 = 2^n v_1$ and $A^n v_2 = 5^n v_2$ ✓
- ☐ D. A^n has eigenvalues $2n, 5n$

Your answer is correct.

The correct answer is: $A^n v_1 = 2^n v_1$ and $A^n v_2 = 5^n v_2$

Question 6

Complete

Mark 2.00 out of 2.00

Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$.

Use **Cayley–Hamilton** to write A^2 in the form $\alpha A + \beta I$, then compute A^4 quickly.

$$(2-\lambda)(3-\lambda) = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$A^2 - 5A + 6I = 0$$

$$A^2 = 5A - 6I$$

$$A^n = \alpha_n A + \beta_n I$$

$$2\alpha_n + \beta_n = 2^n$$

$$3\alpha_n + \beta_n = 3^n$$

$$\alpha_n = 3^{n-2} \cdot 2^n$$

$$\beta_n = 3 \cdot 2^{n-2} - 3^n$$

$$A^{2n} = 3^{n-2} \cdot 2^n A$$

$$A^4 = 5A^3 - 6A^2$$

Comment: